



Use a Linear Prediction Model

Grades 9-12 • SAT / ACT mathematics foundations

Problem-Solving and Data Analysis • Interpret a model prediction and residual.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Interpret a model prediction and residual.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.

Two worked examples

Model 1. A fitted model is $\hat{y} = 7/2x + 18$. At $x = 13$, an observed y is $133/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

Prediction = $7/2 \times 13 + 18 = 127/2$. Residual = $133/2 - 127/2 = 3$. Even a perfect fit does not establish cause.

Model 2. A fitted model is $\hat{y} = 1x + 14$. At $x = 13$, an observed y is 30. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

Prediction = $1 \times 13 + 14 = 27$. Residual = $30 - 27 = 3$. Even a perfect fit does not establish cause.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A fitted model is $\hat{y} = 3/2x + 17$. At $x = 9$, an observed y is $67/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

2. A fitted model is $\hat{y} = 2x + 19$. At $x = 11$, an observed y is 44. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

3. A fitted model is $\hat{y} = 2x + 12$. At $x = 8$, an observed y is 31. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A fitted model is $\hat{y} = 2x + 19$. At $x = 8$, an observed y is 38. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

5. A fitted model is $\hat{y} = 5/2x + 13$. At $x = 13$, an observed y is $97/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

6. A fitted model is $\hat{y} = 4x + 18$. At $x = 10$, an observed y is 61. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A fitted model is $\hat{y} = 7/2x + 16$. At $x = 12$, an observed y is 61. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

8. A fitted model is $\hat{y} = 2x + 18$. At $x = 9$, an observed y is 39. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

9. A fitted model is $\hat{y} = 2x + 12$. At $x = 13$, an observed y is 41. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A fitted model is $\hat{y} = \frac{3}{2}x + 16$. At $x = 11$, an observed y is $7\frac{1}{2}$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

11. A fitted model is $\hat{y} = 2x + 12$. At $x = 13$, an observed y is 41. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

12. A fitted model is $\hat{y} = 3x + 14$. At $x = 7$, an observed y is 38. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A fitted model is $\hat{y} = 3/2x + 19$. At $x = 9$, an observed y is $71/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

14. A fitted model is $\hat{y} = 3/2x + 19$. At $x = 13$, an observed y is $83/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

15. A fitted model is $\hat{y} = 2x + 13$. At $x = 7$, an observed y is 30. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A fitted model is $\hat{y} = 7/2x + 18$. At $x = 11$, an observed y is $119/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A fitted model is $\hat{y} = 9/2x + 14$. At $x = 7$, an observed y is $97/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

2. A fitted model is $\hat{y} = 7/2x + 16$. At $x = 13$, an observed y is $129/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A fitted model is $\hat{y} = 5/2x + 14$. At $x = 8$, an observed y is 37. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

4. A fitted model is $\hat{y} = 7/2x + 15$. At $x = 8$, an observed y is 46. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A fitted model is $\hat{y} = 7/2x + 16$. At $x = 7$, an observed y is $87/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

6. A fitted model is $\hat{y} = 7/2x + 19$. At $x = 7$, an observed y is $93/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A fitted model is $\hat{y} = 7/2x + 15$. At $x = 10$, an observed y is 53. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

2. A fitted model is $\hat{y} = 7/2x + 18$. At $x = 7$, an observed y is $91/2$. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A fitted model is $\hat{y} = 7/2x + 18$. At $x = 8$, an observed y is 49. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?

4. A fitted model is $\hat{y} = 1x + 16$. At $x = 9$, an observed y is 28. Find the prediction and residual (observed minus predicted). Does a residual of zero prove a causal relationship?



Practice worked solutions

1. Prediction 61/2; residual 3; no

Prediction = $3/2 \times 9 + 17 = 61/2$. Residual = $67/2 - 61/2 = 3$. Even a perfect fit does not establish cause.

2. Prediction 41; residual 3; no

Prediction = $2 \times 11 + 19 = 41$. Residual = $44 - 41 = 3$. Even a perfect fit does not establish cause.

3. Prediction 28; residual 3; no

Prediction = $2 \times 8 + 12 = 28$. Residual = $31 - 28 = 3$. Even a perfect fit does not establish cause.

4. Prediction 35; residual 3; no

Prediction = $2 \times 8 + 19 = 35$. Residual = $38 - 35 = 3$. Even a perfect fit does not establish cause.



Practice worked solutions

5. Prediction 91/2; residual 3; no

Prediction = $5/2 \times 13 + 13 = 91/2$. Residual = $97/2 - 91/2 = 3$. Even a perfect fit does not establish cause.

6. Prediction 58; residual 3; no

Prediction = $4 \times 10 + 18 = 58$. Residual = $61 - 58 = 3$. Even a perfect fit does not establish cause.

7. Prediction 58; residual 3; no

Prediction = $7/2 \times 12 + 16 = 58$. Residual = $61 - 58 = 3$. Even a perfect fit does not establish cause.

8. Prediction 36; residual 3; no

Prediction = $2 \times 9 + 18 = 36$. Residual = $39 - 36 = 3$. Even a perfect fit does not establish cause.

Practice worked solutions

9. Prediction 38; residual 3; no

Prediction = $2 \times 13 + 12 = 38$. Residual = $41 - 38 = 3$. Even a perfect fit does not establish cause.

10. Prediction 65/2; residual 3; no

Prediction = $3/2 \times 11 + 16 = 65/2$. Residual = $71/2 - 65/2 = 3$. Even a perfect fit does not establish cause.

11. Prediction 38; residual 3; no

Prediction = $2 \times 13 + 12 = 38$. Residual = $41 - 38 = 3$. Even a perfect fit does not establish cause.

12. Prediction 35; residual 3; no

Prediction = $3 \times 7 + 14 = 35$. Residual = $38 - 35 = 3$. Even a perfect fit does not establish cause.

Practice worked solutions

13. Prediction 65/2; residual 3; no

Prediction = $3/2 \times 9 + 19 = 65/2$. Residual = $71/2 - 65/2 = 3$. Even a perfect fit does not establish cause.

14. Prediction 77/2; residual 3; no

Prediction = $3/2 \times 13 + 19 = 77/2$. Residual = $83/2 - 77/2 = 3$. Even a perfect fit does not establish cause.

15. Prediction 27; residual 3; no

Prediction = $2 \times 7 + 13 = 27$. Residual = $30 - 27 = 3$. Even a perfect fit does not establish cause.

16. Prediction 113/2; residual 3; no

Prediction = $7/2 \times 11 + 18 = 113/2$. Residual = $119/2 - 113/2 = 3$. Even a perfect fit does not establish cause.

Transfer worked solutions

1. Prediction 91/2; residual 3; no

Prediction = $9/2 \times 7 + 14 = 91/2$. Residual = $97/2 - 91/2 = 3$. Even a perfect fit does not establish cause.

2. Prediction 123/2; residual 3; no

Prediction = $7/2 \times 13 + 16 = 123/2$. Residual = $129/2 - 123/2 = 3$. Even a perfect fit does not establish cause.

3. Prediction 34; residual 3; no

Prediction = $5/2 \times 8 + 14 = 34$. Residual = $37 - 34 = 3$. Even a perfect fit does not establish cause.

4. Prediction 43; residual 3; no

Prediction = $7/2 \times 8 + 15 = 43$. Residual = $46 - 43 = 3$. Even a perfect fit does not establish cause.

Transfer worked solutions

5. Prediction $81/2$; residual 3; no

Prediction = $7/2 \times 7 + 16 = 81/2$. Residual = $87/2 - 81/2 = 3$. Even a perfect fit does not establish cause.

6. Prediction $87/2$; residual 3; no

Prediction = $7/2 \times 7 + 19 = 87/2$. Residual = $93/2 - 87/2 = 3$. Even a perfect fit does not establish cause.

Exit-ticket worked solutions

1. Prediction 50; residual 3; no

Prediction = $7/2 \times 10 + 15 = 50$. Residual = $53 - 50 = 3$. Even a perfect fit does not establish cause.

2. Prediction 85/2; residual 3; no

Prediction = $7/2 \times 7 + 18 = 85/2$. Residual = $91/2 - 85/2 = 3$. Even a perfect fit does not establish cause.

3. Prediction 46; residual 3; no

Prediction = $7/2 \times 8 + 18 = 46$. Residual = $49 - 46 = 3$. Even a perfect fit does not establish cause.

4. Prediction 25; residual 3; no

Prediction = $1 \times 9 + 16 = 25$. Residual = $28 - 25 = 3$. Even a perfect fit does not establish cause.

Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Interpret a model prediction and residual. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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